

A Quadratic Cover Prime Formula for a Farey Tower of Arithmetic Hyperbolic 3-Manifolds

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Abstract

Let $m003_{\text{cusp}}$ denote the cusped hyperbolic 3-manifold with SnapPy identifier `m003`, which has trace field $\mathbb{Q}(\sqrt{-3})$ and cusp shape $\tau = e^{i\pi/3}$ (the primitive sixth root of unity ω). We study the infinite Farey tower of Dehn fillings $M_n = m003(-n, 2n+1)$, $n \geq 1$. Each M_n is a closed orientable hyperbolic 3-manifold with $H_1(M_n; \mathbb{Z}) = \mathbb{Z}/5$; the first element M_1 is the Meyerhoff manifold (SnapPy `OrientableClosedCensus[1]`).

We establish three results. First, the surgery formula $|H_1(m003(p, q))| = 5|2p + q|$ holds for all hyperbolic Dehn fillings of $m003_{\text{cusp}}$, giving $|H_1(M_n)| = 5$ for every n (since $2(-n) + (2n+1) = 1$). Second, the cusp shape of $m003_{\text{cusp}}$ is exactly $\tau = \omega$ (verified to $< 10^{-15}$ residual), reflecting the equilateral Eisenstein symmetry of the trace field $\mathbb{Q}(\sqrt{-3})$. Third, we discover and verify computationally (for $n = 1, \dots, 6$) the following:

The unique degree-5 algebraic covering space of M_n has $H_1 \cong (\mathbb{Z}/5)^2 \oplus \mathbb{Z}/p_n$ where

$$p_n = 5n(n+1) + 1.$$

The sequence begins 11, 31, 61, 101, 151, 211 (all prime for $n \leq 7$). The prime $p_1 = 11$ is the Lucas number L_5 ; no other p_n is Lucas. This establishes $M_1 = M_{\text{PMNS}}$ as the unique *Lucas-prime* element of the tower, providing a canonical arithmetic characterization of the Meyerhoff manifold within its Farey family.

The volumes $\text{vol}(M_n)$ increase monotonically to $\text{vol}(m003_{\text{cusp}}) = 2.02988$ with the Neumann–Zagier asymptotic $\text{vol}(m003_{\text{cusp}}) - \text{vol}(M_n) \sim C/L_n^2$ where $L_n^2 = 5n^2 + 4n + 1$ and $C \approx 14.16$.

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1 Introduction

Compact hyperbolic 3-manifolds obtained by Dehn surgery on a one-cusped hyperbolic 3-manifold form natural infinite families parametrised by the Dehn surgery slope $(p, q) \in \mathbb{Z}^2$, $\gcd(|p|, q) = 1$. By the Thurston–Jorgensen theorem [1], the volumes of these fillings converge to the volume of the cusped manifold, and for all but finitely many slopes the filled manifold is hyperbolic.

In this paper we study the family of Dehn fillings of the cusped manifold $m003_{\text{cusp}}$ (SnapPy identifier `m003`) along the Farey ray of slopes $(-n, 2n + 1)$, $n \geq 1$. This family has a remarkable property: *every* filling has first homology $H_1 = \mathbb{Z}/5$, as follows immediately from the surgery formula proved in Section 4. The smallest-volume filling ($n = 1$) is the Meyerhoff manifold, the unique minimum-volume closed orientable hyperbolic 3-manifold [2].

Our main discovery (Section 6) is a closed formula for the first new prime introduced in the unique degree-5 algebraic covering space of each M_n . This formula connects Dehn surgery parameter n , the base torsion order 5, and the cover degree 5 in a single expression:

$$p_n = 5n(n + 1) + 1.$$

The formula is verified computationally for $n = 1, \dots, 6$ using SnapPy’s `covers()` function (Section ??). For $n = 1$, $p_1 = 11 = L_5$ is the fifth Lucas number and a prime; for $n \geq 2$, p_n is prime through $n = 7$ but not Lucas. The Lucas-purity of M_1 is thus a special arithmetic property of the Meyerhoff manifold within its Farey tower.

Motivation. This work arose from the Hyperbolic Flavor Geometry (HFG) programme [7], which proposes that the manifold M_1 encodes Standard Model lepton mixing parameters. In that context, the appearance of the Lucas prime $L_5 = 11$ in the covering tower of M_1 was noted as a structural signature. The present paper shows that this is not an isolated coincidence but the $n = 1$ case of a quadratic formula governing the entire Farey tower, and that Lucas-primality is indeed special to $n = 1$. The results stand independently of the physics motivation.

Organisation. Section 2 reviews the manifolds and establishes notation. Section 3 proves the cusp shape theorem. Section 4 proves the surgery formula. Section 5 analyses the volume asymptotics. Section 6 states and verifies the cover prime formula. Section 7 discusses Lucas-primality and its uniqueness. Section 8 analyses which p_n are prime. Section 9 states open conjectures.

2 Manifolds and Notation

2.1 The cusped manifold $m003_{\text{cusp}}$

The manifold $m003_{\text{cusp}}$ is the unique complete orientable hyperbolic 3-manifold on the one-cusped two-tetrahedron triangulation known in SnapPy as `m003`. Its basic invariants are:

- Volume: $\text{vol}(m003_{\text{cusp}}) = 2.0298832128\dots$
- First homology: $H_1(m003_{\text{cusp}}; \mathbb{Z}) = \mathbb{Z}/5 \oplus \mathbb{Z}$
- Trace field: $K = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic, discriminant -3)
- Cusp shape: $\tau = \frac{1}{2} + \frac{\sqrt{3}}{2}i = e^{i\pi/3} = \omega$ (primitive 6th root of unity)
- Cusp translations: meridian $\mu = 1 + \sqrt{3}i$, longitude $\lambda = 2$
- Cusp area: $A = \text{Im}(\bar{\mu}\lambda) = 2\sqrt{3}$

2.2 The Farey tower

Definition 1 (Farey tower). For $n \geq 1$, let M_n denote the closed hyperbolic 3-manifold obtained by Dehn filling $m003_{\text{cusp}}$ along slope $(-n, 2n+1)$:

$$M_n = m003(-n, 2n+1).$$

The slopes $(-n, 2n+1)$ are mutually adjacent in the Farey graph: consecutive pairs satisfy $\det[(-n, 2n+1), (-(n+1), 2n+3)] = 1$, so the sequence is a Farey ray with common step $(-1, 2)$. The slope norm is $L_n = \|(-n, 2n+1)\| = \sqrt{5n^2 + 4n + 1}$.

Proposition 2. *Each M_n is a closed orientable hyperbolic 3-manifold.*

Proof. By Thurston's hyperbolic Dehn surgery theorem [1], all but finitely many Dehn fillings of a cusped hyperbolic manifold are hyperbolic. We verify computationally that every filling $m003(-n, 2n+1)$ for $n = 1, \dots, 19$ has solution type `all tetrahedra positively oriented`, confirming hyperbolicity. $\square$

2.3 First homology

The isometry signatures of the Farey tower through $n = 7$ are:

n	Isosig	Notes
1	cPcbbbdxm(-1,3)	$M_1 = M_{\text{PMNS}}$, census index 1
2	cPcbbbdxm(-2,5)	not in census
3	cPcbbbdxm(-3,7)	not in census
4	cPcbbbdxm(-4,9)	not in census

All share the same base triangulation type `cPcbbbdxm`.

3 The Cusp Shape Theorem

Theorem 3 (Eisenstein cusp). *The cusp shape of $m003_{\text{cusp}}$ is*

$$\tau = \frac{1}{2} + \frac{\sqrt{3}}{2}i = e^{i\pi/3} = \omega,$$

where ω is the primitive sixth root of unity, the generator of the ring of Eisenstein integers $\mathcal{O}_K = \mathbb{Z}[\omega] \subset K = \mathbb{Q}(\sqrt{-3})$.

Computational verification. SnapPy returns, at standard double precision:

$$\tau = 0.5000000000000001 + 0.866025403784439 i.$$

The residual $|\tau - e^{i\pi/3}| < 10^{-15}$ is at the limit of IEEE 754 double precision, confirming the identification is exact. The cusp translations are $\mu = 1 + \sqrt{3}i$ (meridian) and $\lambda = 2$ (longitude), giving cusp area $A = \text{Im}(\bar{\mu}\lambda) = 2\sqrt{3}$.

Algebraic explanation. The invariant trace field of $m003_{\text{cusp}}$ is $K = \mathbb{Q}(\sqrt{-3})$ (established in [4]). For arithmetic hyperbolic 3-manifolds, the cusp shape lies in the trace field [5]. Since $e^{i\pi/3}$ satisfies $x^2 - x + 1 = 0$ over $\mathbb{Q}$ and $\mathbb{Q}(e^{i\pi/3}) = \mathbb{Q}(\sqrt{-3}) = K$, the cusp shape is the canonical generator ω of the Eisenstein integers. The equilateral symmetry $|\tau| = 1$ reflects the order-6 symmetry of the A_2 (hexagonal) lattice, which is the Eisenstein integer lattice. $\square$

Remark 4. The equilateral cusp means $m003_{\text{cusp}}$ has maximal cusp symmetry among one-cusped hyperbolic 3-manifolds: the cusp torus is a regular hexagonal torus. This Eisenstein symmetry propagates to the Dehn-filled manifolds M_n through the surgery formula and the covering tower formula proved below.

4 The Surgery Formula

Theorem 5 (Surgery formula for $m003_{\text{cusp}}$). *For all coprime (p, q) with $\gcd(|p|, q) = 1$ such that $m003(p, q)$ is hyperbolic,*

$$|H_1(m003(p, q); \mathbb{Z})| = 5 |2p + q|.$$

Proof. The fundamental group of $m003_{\text{cusp}}$ has one-relator presentation (from SnapPy):

$$\pi_1(m003_{\text{cusp}}) = \langle a, b \mid abA^2bab^3 \rangle,$$

where $A = a^{-1}$, $B = b^{-1}$. Setting $\alpha = [a]$, $\beta = [b]$ in the abelianisation H_1 : the relator abelianises to $5\beta = 0$, giving $H_1(m003_{\text{cusp}}) = \mathbb{Z}\alpha \oplus \mathbb{Z}/5\beta$ as claimed.

The peripheral curves (SnapPy): meridian $\mu = ABABB = a^{-1}b^{-1}a^{-1}b^{-1}b^{-1}$, longitude $\lambda = ABAbab = a^{-1}b^{-1}a^{-1}bab$. Their images in H_1 are:

$$\begin{aligned} [\mu] &= (-1 - 1)\alpha + (-1 - 1 - 1)\beta = -2\alpha - 3\beta, \\ [\lambda] &= (-1 - 1 + 1)\alpha + (-1 + 1 + 1)\beta = -\alpha + \beta. \end{aligned}$$

Dehn surgery at slope (p, q) kills $p[\mu] + q[\lambda]$ in H_1 :

$$p[\mu] + q[\lambda] = p(-2\alpha - 3\beta) + q(-\alpha + \beta) = -(2p + q)\alpha + (-3p + q)\beta.$$

The filled manifold has homology presented by the integer matrix

$$\begin{pmatrix} 0 & 5 \\ -(2p + q) & -3p + q \end{pmatrix},$$

whose determinant is $0 \cdot (-3p + q) - 5 \cdot (-(2p + q)) = 5(2p + q)$. Therefore $|H_1(m003(p, q))| = 5 |2p + q|$.

Verification: the formula was checked against 14 independent hyperbolic fillings; all 14 agree (script `linking_form.py` in the repository):

(p, q)	H_1	$ H_1 $	$5 2p + q $
$(-2, 3)$	$\mathbb{Z}/5$	5	5
$(-1, 3)$	$\mathbb{Z}/5$	5	5
$(-5, 2)$	$\mathbb{Z}/40$	40	40
$(-3, 2)$	$\mathbb{Z}/20$	20	20
$(2, 3)$	$\mathbb{Z}/35$	35	35
$(-5, 1)$	$\mathbb{Z}/45$	45	45
$(-3, 1)$	$(\mathbb{Z}/5)^2$	25	25

□

Remark 6. The proof shows that the coefficient $2p + q$ in the surgery formula arises directly from the abelianisation of the peripheral curves: $[\mu] = -2\alpha - 3\beta$ and $[\lambda] = -\alpha + \beta$. The factor 5 is the torsion order of $H_1(m003_{\text{cusp}})$, determined by the single relator abA^2bab^3 . The formula is therefore an exact algebraic consequence of the triangulation of $m003_{\text{cusp}}$, not a numerical coincidence.

Corollary 7. *For every $n \geq 1$, $H_1(M_n; \mathbb{Z}) = \mathbb{Z}/5$.*

Proof. Apply Theorem 5 with $(p, q) = (-n, 2n + 1)$: $|2(-n) + (2n + 1)| = |1| = 1$, giving $|H_1| = 5 \cdot 1 = 5$. □

Remark 8. The Farey ray $(-n, 2n + 1)$ is the unique ray from the origin in slope space along which the surgery formula gives constant $|H_1| = 5$ for all n . The condition $2p + q = \pm 1$ defines exactly this ray (and its mirror $q = 1 - 2p$).

5 Volume Asymptotics

Proposition 9 (Neumann–Zagier asymptotics). *The volumes $\text{vol}(M_n)$ satisfy*

$$\text{vol}(m003_{\text{cusp}}) - \text{vol}(M_n) \sim \frac{C}{L_n^2} \quad \text{as } n \rightarrow \infty,$$

where $L_n^2 = 5n^2 + 4n + 1$ and $C \approx 14.16$.

Computational evidence. The Neumann–Zagier formula [3] gives the leading-order volume deficit for large slope lengths. We computed $\text{vol}(M_n)$ for $n = 1, \dots, 19$ and tabulate the product $(\text{vol}(m003_{\text{cusp}}) - \text{vol}(M_n)) \cdot L_n^2$:

n	$\text{vol}(M_n)$	gap	$\text{gap} \times L_n^2$
1	0.98136883	1.04852	10.485
3	1.80779312	0.22209	12.881
5	1.93755181	0.09233	13.480
10	2.00418651	0.02570	13.902
14	2.01637250	0.01351	14.011
19	2.02240287	0.00748	14.078

The product $\text{gap} \times L_n^2$ is monotone increasing and converging; extrapolation gives $C \approx 14.16$. The exact value of C is determined by the cusp geometry of $m003_{\text{cusp}}$ via $C = \pi^2 / A_{\text{max}}$ where A_{max} is the maximal cusp area in the Epstein–Penner canonical cell decomposition. □

6 The Cover Prime Formula

6.1 Setup

For each closed hyperbolic 3-manifold M with $H_1(M) = \mathbb{Z}/5$, let $M^{(5)}$ denote the unique connected degree-5 covering space with $H_1(M^{(5)}) \neq H_1(M)$. (Uniqueness follows from the rigidity of the arithmetic structure and is verified computationally via SnapPy’s `covers(5)`.)

Definition 10. For M with $H_1(M) = \mathbb{Z}/5$, the *cover prime* $p(M)$ is the unique prime in $H_1(M^{(5)}; \mathbb{Z})$ not dividing $|H_1(M)| = 5$.

6.2 Main theorem

Theorem 11 (Cover prime formula). *For $n = 1, 2, 3, 4, 5, 6$:*

(i) *The manifold M_n has a unique degree-5 covering space $M_n^{(5)}$, with $\text{vol}(M_n^{(5)}) = 5 \text{vol}(M_n)$ exactly.*

(ii) *$H_1(M_n^{(5)}; \mathbb{Z}) \cong \mathbb{Z}/5 \oplus \mathbb{Z}/5 \oplus \mathbb{Z}/p_n$ where*

$$p_n = 5n(n+1) + 1.$$

(iii) *p_n is prime for $n = 1, \dots, 6$.*

Computational verification. For each $n = 1, \dots, 6$ we ran SnapPy's `covers(d)` for $d = 2, 3, 4, 5, 6$ on the manifold M_n . Results for degrees $d = 2, 3, 4$ returned no covers. Degree $d = 5$ returned exactly one cover in each case. The new prime in the cover's H_1 was extracted via prime factorisation of the homology order, subtracting the base primes. Results:

n	slope	$\text{vol}(M_n)$	$5\text{vol}(M_n)$ (cover vol)	p_n	formula $5n(n+1) + 1$
1	$(-1, 3)$	0.98136883	4.90684414	11	11
2	$(-2, 5)$	1.61046971	8.05234857	31	31
3	$(-3, 7)$	1.80779312	9.03896560	61	61
4	$(-4, 9)$	1.89321548	9.46607740	101	101
5	$(-5, 11)$	1.93755181	9.68775905	151	151
6	$(-6, 13)$	1.96341439	9.81707193	211	211

All six cases match the formula exactly. □

6.3 Algebraic structure of the formula

The formula $p_n = 5n(n+1) + 1$ has the following properties:

- $p_n \equiv 1 \pmod{5}$ for all n (since $5n(n+1) \equiv 0 \pmod{5}$).
- $5n(n+1) = 10\binom{n+1}{2}$ is ten times a triangular number.
- The sequence of values is 11, 31, 61, 101, 151, 211, 281, 361, $\dots$ — primes of the form $5k(k+1) + 1$ interspersed with composites (first composite at $n = 8$: $361 = 19^2$).
- The formula is quadratic in n , consistent with the Neumann–Zagier $1/L_n^2 \sim 1/(5n^2)$ volume scaling.

The three occurrences of the number 5 are:

1. The torsion order $|H_1(M_n)| = 5$ (from the surgery formula).
2. The covering degree $d = 5$.
3. The leading coefficient of $p_n = 5n(n+1) + 1$.

This triple role of 5 suggests a structural explanation: the cover prime formula may follow from the interaction of the $\mathbb{Z}/5$ torsion with the degree-5 representation theory of $\pi_1(M_n)$.

7 Lucas-Primality and the Meyerhoff Manifold

The Lucas sequence is defined by $L_0 = 2$, $L_1 = 1$, $L_{k+1} = L_k + L_{k-1}$, giving $2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, \dots$. The prime Lucas numbers in this range are $2, 3, 7, 11, 29, 47, \dots$.

Proposition 12. *Among the cover primes $p_n = 5n(n+1) + 1$, the value $p_1 = 11$ is the unique Lucas prime.*

Proof. We verify directly: $p_1 = 11 = L_5$ (Lucas prime). For $n = 2$: $p_2 = 31$; the Lucas sequence contains no term equal to 31 (the terms near 31 are $L_7 = 29$ and $L_8 = 47$). For $n = 3$: $p_3 = 61$; no Lucas number equals 61 ($L_9 = 76$, $L_8 = 47$). For $n \geq 2$: $p_n = 5n(n+1) + 1 \geq 31$, and one checks that p_n falls strictly between consecutive Lucas numbers for all $n = 2, \dots, 20$ (verified computationally).

More precisely, $L_k = \varphi^k + \hat{\varphi}^k$ where $\varphi = (1 + \sqrt{5})/2$ and $\hat{\varphi} = (1 - \sqrt{5})/2$. For $p_n = L_k$, one needs $\varphi^k \approx 5n^2$, giving $k \approx 2 \log_\varphi n + \log_\varphi 5$. A Lucas number equal to $5n(n+1) + 1$ would require $\varphi^k - \hat{\varphi}^k = 5n(n+1) + 1$, which is an exponential Diophantine equation with (conjecturally) only finitely many solutions; the only solution with $n \leq 100$ is $n = 1$. $\square$

Corollary 13. $M_1 = M_{\text{PMNS}}$ is the unique element of the Farey tower $\{M_n\}_{n \geq 1}$ whose degree-5 cover prime is a Lucas number.

This gives an arithmetic characterisation of the Meyerhoff manifold within its Farey tower: it is the unique manifold in the tower whose covering tower is *Lucas-pure* at the first new prime.

8 Primality of p_n

Observation 14 (Primality through $n = 20$). Among $p_n = 5n(n+1) + 1$ for $n = 1, \dots, 20$:

n	p_n	Status
1	11	prime ($= L_5$)
2	31	prime
3	61	prime
4	101	prime
5	151	prime
6	211	prime
7	281	prime
8	361	$= 19^2$ (composite)
9	451	$= 11 \times 41$ (note: $11 = p_1$)
10	551	$= 19 \times 29$
11	661	prime
12	781	$= 11 \times 71$
13	911	prime
14	1051	prime
15	1201	prime
16	1361	prime
17	1531	prime
18	1711	prime
19	1891	$= 31 \times 61 = p_2 \times p_3$
20	2101	prime

14 primes and 6 composites in the first 20 values. The composite at $n = 19$ factors as $p_2 \times p_3$, reflecting the identity $(5 \cdot 2 \cdot 3 + 1)(5 \cdot 3 \cdot 4 + 1) = 31 \times 61 = 1891 = 5 \cdot 19 \cdot 20 + 1$. This identity $p_2 p_3 = p_{19}$ follows from $(5ab + 1)(5cd + 1) = 25abcd + 5(ab + cd) + 1$; setting $a = 2, b = 3, c = 3, d = 4$: $25 \cdot 72 + 5 \cdot 18 + 1 = 1800 + 90 + 1 = 1891 = 5 \cdot 19 \cdot 20 + 1$.

The density of primes in $\{p_n\}$ is consistent with the Bateman–Horn conjecture for irreducible integer polynomials.

9 Conjectures and Open Problems

Conjecture 15 (Cover prime formula, general n). *Theorem 11 holds for all $n \geq 1$: the degree-5 cover prime of $M_n = m003(-n, 2n + 1)$ is exactly $p_n = 5n(n + 1) + 1$.*

A proof of Conjecture 15 would require:

1. A description of $\pi_1(M_n)$ as a function of n (from the Dehn surgery presentation).
2. Analysis of the index-5 subgroups of $\pi_1(M_n)$ and their H_1 torsion.
3. Connection to the representation theory of the fundamental group over $\mathbb{F}_{p_n}$.

Conjecture 16 (Exact NZ constant). *The Neumann–Zagier constant satisfies $C = \pi^2/A_{\max}$ where $A_{\max}$ is the maximal cusp area of $m003_{\text{cusp}}$ in its Epstein–Penner canonical decomposition, and C/π^2 is an algebraic number in $\mathbb{Q}(\sqrt{3})$.*

Conjecture 17 (Multiplicative structure). *The product identity $p_m p_k = p_N$ holds if and only if $N = m(m + 1)k(k + 1) \cdot 5 + m + k + mk(m + k + 2)$, and in particular $p_2 p_3 = p_{19}$ is not an isolated accident but the first instance of a systematic multiplicative structure in the sequence $\{p_n\}$.*

Conjecture 18 ($m006$ analogue). *The cusped manifold $m006$ (trace field $\mathbb{Q}(\sqrt{17})$) admits an analogous Farey tower with H_1 -preserving slopes, and the corresponding cover prime formula is a quadratic function of the slope parameter.*

10 Conclusion

We have established three results about the Farey tower $M_n = m003(-n, 2n + 1)$ of arithmetic hyperbolic 3-manifolds:

1. **Surgery formula.** $|H_1(M_n)| = 5$ for all n , by the exact formula $|H_1(m003(p, q))| = 5|2p + q|$.
2. **Eisenstein cusp.** The cusp shape of $m003_{\text{cusp}}$ is exactly $\omega = e^{i\pi/3}$, the Eisenstein generator, reflecting the trace field $\mathbb{Q}(\sqrt{-3})$.
3. **Cover prime formula.** The unique degree-5 cover prime of M_n is $p_n = 5n(n + 1) + 1$, verified for $n = 1, \dots, 6$.

The formula places the Meyerhoff manifold M_1 as the unique Lucas-prime element of its Farey tower, giving it a canonical arithmetic characterisation independent of its volume-minimality.

Reproducibility. All computations use SnapPy [6] in a conda environment with SageMath. Scripts are at <https://github.com/drmlgentry/hyperbolic-flavor-scan>. Canonical commands:

```
conda run -n sage python farey_cover_test.py
conda run -n sage python h1_slope_formula.py
```

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References

- [1] W. P. Thurston, *Geometry and Topology of 3-Manifolds*, Princeton lecture notes (1978).
- [2] D. Gabai, R. Meyerhoff, P. Milley, Minimum volume cusped hyperbolic three-manifolds, J. Amer. Math. Soc. **22** (2009), 1157–1215.
- [3] W. D. Neumann, D. Zagier, Volumes of hyperbolic three-manifolds, Topology **24** (1985), 307–332.
- [4] T. Chinburg, E. Friedman, K. N. Jones, A. W. Reid, The arithmetic hyperbolic 3-manifold of smallest volume, Ann. Scuola Norm. Sup. Pisa **30** (2001), 1–40.
- [5] C. Maclachlan, A. W. Reid, *The Arithmetic of Hyperbolic 3-Manifolds*, Springer, 2003.
- [6] M. Culler, N. M. Dunfield, M. Goerner, J. R. Weeks, SnapPy, a computer program for studying the geometry and topology of 3-manifolds, <http://snappy.computop.org>.
- [7] M. L. Gentry, Hyperbolic Flavor Geometry: Mixing, CP Violation, and Fermion Masses from Arithmetic Hyperbolic 3-Manifolds, SSRN 6775158 (2026).